

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Seasoned backend engineer with 9+ years of hands-on experience designing and scaling high-throughput, low-latency systems using Scala, Akka, Kafka, and cloud-native tools across AWS and GCP. Specialized in real-time stream processing, distributed architectures, and building resilient, event-driven systems powering digital advertising, analytics, and data pipelines. Proven record of ownership and impact in remote-first, cross-functional environments. Passionate about AdTech, functional programming, and solving complex problems at scale.

TECHNICAL SKILLS

- **Languages:** Scala, Java, Python, JavaScript, TypeScript, SQL, NoSQL, Shell
- **Frameworks & Tools:** Akka, Play Framework, Spark, Flink, Kafka, RabbitMQ, Redis, Aerospike
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis
- **Cloud & DevOps:** AWS (S3, RDS, Lambda, ECS, CloudWatch), GCP (BigQuery, Pub/Sub), Docker, Kubernetes, GitHub Actions, Jenkins
- **Monitoring:** Grafana, Prometheus, ELK, New Relic
- **Other:** CI/CD, REST APIs, WebSockets, OpenRTB, bidstream parsing, GDPR compliance

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Led the design and implementation of high-throughput streaming services using Scala, Kafka, and Akka, handling over 100K+ concurrent ad impressions per second.
- Built and maintained batch and real-time data pipelines across AWS and GCP, integrating S3, BigQuery, and Cloud Functions for automated data ingestion and transformation.
- Refactored monolithic legacy systems into decoupled microservices, achieving 40% performance improvement and increased scalability in production load testing.
- Integrated Redis and Aerospike as in-memory data stores for low-latency data retrieval, boosting system responsiveness under burst traffic conditions.
- Collaborated with cross-functional teams to develop scalable APIs for real-time bidding (RTB), ensuring compliance with OpenRTB standards.
- Led the introduction of observability tooling including Prometheus and Grafana, establishing full-stack alerting and system health visibility across services.

- Worked with legal and compliance to embed GDPR-compliant data anonymization and deletion workflows into backend pipelines.
- Mentored 4 junior engineers, introducing code review practices and functional programming paradigms to increase team code quality and throughput.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Developed distributed campaign tracking systems using Scala and Spark, analyzing over 50 million user actions per day for AdTech clients.
- Designed concurrent processing workflows using Akka Streams and Futures, minimizing latency and ensuring back-pressure handling in message-heavy systems.
- Created integration bridges with major AdTech vendors via OpenRTB, ensuring compliant bidstream management and fast auction responses.
- Tuned Postgres and Redis databases for optimal indexing and caching strategies, reducing query latency by 60% in production.
- Implemented scalable event consumers with Kafka and Protobuf, supporting schema evolution and partitioned message handling at scale.
- Deployed backend services in AWS ECS and Lambda, leveraging IaC with CloudFormation and automated rollbacks via CI/CD pipelines.
- Introduced centralized logging using ELK stack, allowing product and QA teams to trace issues and improve debugging effectiveness.
- Championed the transition from Jenkins to GitHub Actions for streamlined builds, increasing delivery velocity and reducing pipeline failures.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Built RESTful backend services with Scala and Play Framework, powering marketing automation platforms for real-time targeting and segmentation.
- Designed SQL and NoSQL schemas (PostgreSQL, MongoDB) tailored for large-scale analytics workloads and multi-tenant architectures.
- Delivered async job scheduling with Akka actors, ensuring reliable message delivery and retry policies for campaign dispatch systems.
- Maintained Kafka cluster pipelines for time-series metrics, integrating with Spark batch jobs for dashboard and reporting insights.
- Applied memory tuning and JVM optimizations to support backend services with 99.99% uptime during peak loads.
- Developed admin dashboards with metrics visualization powered by Grafana and WebSocket streaming for real-time updates.
- Integrated third-party APIs for user data enrichment, with rate-limiting and backoff retry strategies for resilience.
- Participated in agile ceremonies, user story breakdown, and collaborative sprint planning in remote team setups.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Collaborated on an analytics backend using Scala, Akka HTTP, and PostgreSQL, processing mobile SDK telemetry data in near real-time.
- Created scalable job queues and metrics aggregation workers for distributed environments using RabbitMQ and Celery.
- Built dashboards for marketing clients with user segmentation and A/B testing metrics visualization.
- Developed internal tools for log ingestion, automated error classification, and tracing anomalies in backend systems.
- Designed ETL workflows with cron-triggered scripts and scheduled Spark batch jobs.
- Conducted codebase audits and rewrites to adopt consistent functional programming style across services.
- Supported dev-ops with Dockerized builds and integration testing pipelines on GitLab CI.
- Documented APIs and internal developer guides, enabling faster onboarding and clearer technical knowledge sharing.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Assisted in the development of an event-driven ticketing backend system with Scala and PostgreSQL.
- Implemented test harnesses for APIs using ScalaTest and HTTP mocking utilities to improve regression coverage.
- Wrote internal command-line tools for log parsing and time-series visualization during incident postmortems.
- Improved SQL query efficiency through indexing and optimized joins, reducing report generation times significantly.
- Participated in the deployment of microservices using Docker and supervisor scripts on bare metal.
- Built REST endpoints for order and payment management using Play Framework.
- Integrated third-party payment gateway APIs and ensured correct transaction reconciliation workflows.
- Supported unit and integration test cycles, reporting and resolving bugs in coordination with senior developers.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- Advanced Scala and Functional Programming – Rock the JVM (2024)
- AWS Certified Solutions Architect – Associate (2023)

- Google Cloud Professional Data Engineer (2023)
 - Apache Kafka Fundamentals – Confluent (2023)
 - GDPR Compliance for Engineers – DataCamp (2023)
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PROJECTS

- **Real-Time Ad Exchange Engine**
Built a Scala-based ad exchange using Akka and Kafka, processing up to 150K bid requests/sec with OpenRTB compliance. Deployed in AWS ECS with metrics via Prometheus.
- **Distributed Bidstream Processor**
Designed a Flink-based streaming data pipeline for high-frequency bid data ingestion and enrichment. Achieved end-to-end latency under 300ms for 90th percentile.
- **Cross-Cloud Data Lake Integration**
Developed batch pipelines in Spark to unify and normalize ad performance data from AWS S3 and GCP BigQuery. Enabled unified analytics layer for reporting and forecasting.